

Recent results on nonconvex Frank-Wolfe splitting algorithms

Frank-Wolfe, Moreau, and more

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Frank-Wolfe Splitting Algorithms

1. Background
2. History & Motivation
3. Approach
4. Analysis
5. Conclusion

Optimization intro

Optimization in a nutshell ($\mathcal{H} = \mathbb{R}^n$ or any real Hilbert space)

- **Objective function** $f: \mathcal{H} \rightarrow \mathbb{R} \cup \{+\infty\}$.
e.g., data fidelity in ML, energy efficiency, profit, statistical error, ...
- An “optimal” $x \in \mathcal{H}$ makes $f(x)$ the smallest or largest
e.g., **minimize** error, **maximize** efficiency

$$\underset{x \in \mathcal{H}}{\text{minimize}} \ f(x)$$

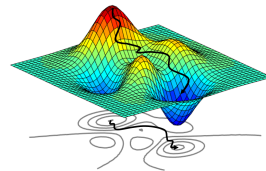


image: towardsdatascience.com

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- **Objective function** $f: \mathcal{H} \rightarrow \mathbb{R} \cup \{+\infty\}$.
e.g., data fidelity in ML, energy efficiency, profit, statistical error, ...
- **Constraint set(s)** $\mathcal{C} \subset \mathcal{H}$
e.g., $\mathbb{R}_+^N$, $\mathbb{S}_+^N$, hypercube, ℓ_p ball, solution set of an inverse problem, ...
- An “optimal” $x \in \mathcal{C}$ makes $f(x)$ the smallest or largest
e.g., minimize error, maximize efficiency

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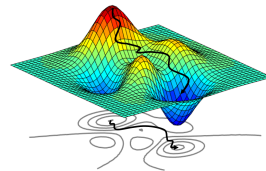
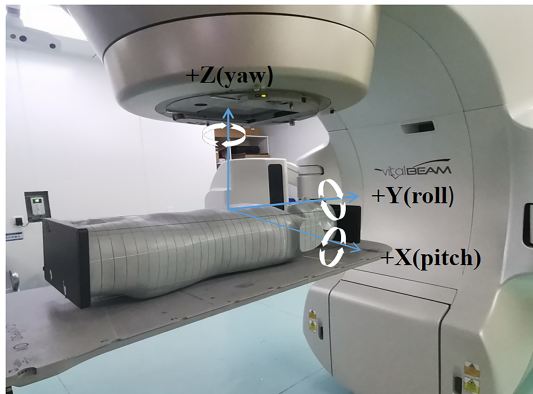


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Modeling via optimization



[Torelli et al., *Med. Phys.*, 2023]

image: [Fu et al., *Tech. Cancer Res. Treatment*, 2023]

Modeling via optimization

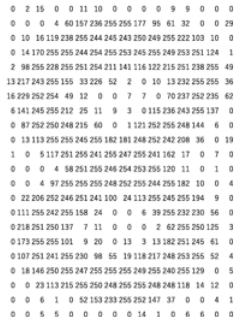
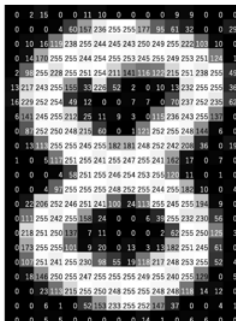


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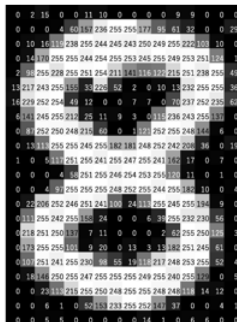
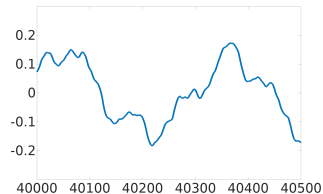
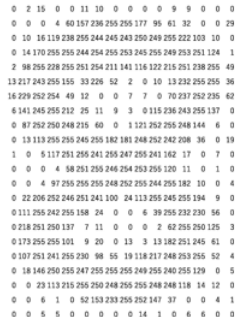


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Modeling via optimization

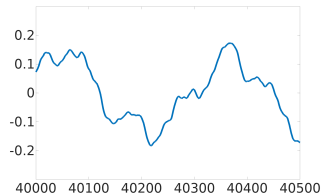
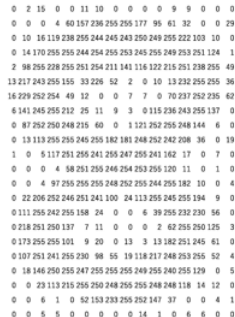
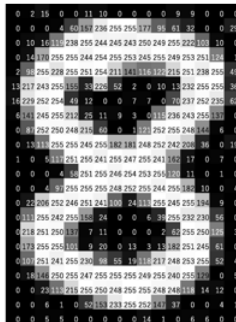


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High-dimensional problems (computing $\nabla^2 f$ is costly)

⇒ use first-order methods (use evaluations of ∇f , prox_f , and/or f)

Modeling via optimization

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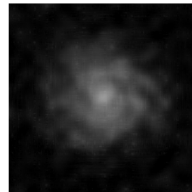
Original



Observation



Recovered stars



Recovered galaxy

[Combettes & ZW., *SIAM J. Imaging Sci.*, 2022]

A few common constraints

Set name $C \subset \mathcal{H}$ and space $\mathcal{H}$	Constraint(s) on $x \in C$	Application Examples
ℓ_p ball ($p \geq 1$) $\mathcal{H} = \mathbb{R}^n$	$\sqrt[p]{\sum_{i \in I} x_i^p} \leq \gamma$ ($\gamma \geq 0$)	Signal processing (e.g., sparse recovery, matrix completion), graph analysis, relaxed combinatorial problems
Hypercube $\mathcal{H} = \mathbb{R}^n$	$\alpha_i \leq x_i \leq \beta_i$ ($\alpha_i, \beta_i \in \mathbb{R}$)	Image and signal processing, recommender system training, graph refinement
Birkhoff Polytope $\mathcal{H} = \mathbb{R}^{n \times n}$	$x_{i,j} \geq 0,$ $\sum_{i \in [n]} x_{i,j} = 1,$ $\sum_{j \in [n]} x_{i,j} = 1$	Graph automorphisms, communication theory, optimal transport, cluster analysis

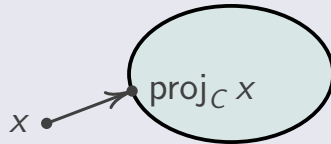
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Polyhedron $\mathcal{H} = \mathbb{R}^n$	$(Ax)_i \leq b_i$ $(A \in \mathbb{R}^{n \times m}, b \in \mathbb{R}^m)$	Optimal transport, production scheduling, inventory control, network flow, computational geometry
Spectrahedron $\mathcal{H} = \mathbb{S}^{n \times n}$	$x \succeq 0$, $\text{trace}(x) = 1$	Covariance estimation, relaxed combinatorial problems, low-rank recovery
Nuclear norm ball $\mathcal{H} = \mathbb{R}^{n \times m}$	$\sum_{i=1}^{\min\{n,m\}} S_{i,i} \leq \gamma$ $(x = USV^T, \gamma \geq 0)$	Graph denoising, recommender system training, cluster analysis, protein interaction prediction, low-rank recovery

A tale of two subroutines

Projection operators

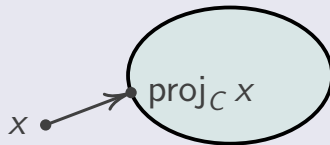
$$\text{proj}_C: x \mapsto \underset{c \in C}{\text{Argmin}} \|x - c\|^2$$



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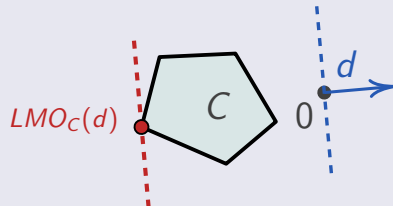
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Linear Minimization Oracles (used in Frank-Wolfe algorithms, AKA “Conditional Gradient” algs.)

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Open source repositories:

proximity-operator.net

[ProximalOperators.jl](#)

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[C. Combettes, Pokutta, '21] + many others:

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If C_i is compact/convex, approximate **LMO** is no slower than **projection**.

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Question:

Is there a compact convex set C such that proj_C requires fewer arithmetic operations than LMO_C ?

Comparison with projections

Comparison with projections

Example: Nuclear norm ball

For $x \in \mathbb{R}^{n \times n}$

$$\|x\|_{nuc} = \sum_{1 \leq i \leq n} \sigma_i(x).$$

For $\beta \geq 0$, $C = \{x \in \mathbb{R}^{n \times n} \mid \|x\|_{nuc} \leq \beta\}$,

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$\text{proj}_C(x)$: requires a **full SVD**!

$(\sigma_1, \dots, \sigma_n, U, V)$, where $x = U \text{diag}(\sigma_1, \dots, \sigma_n) V^T$

Full SVD

$n = 500$: 0.11 sec

$n = 1000$: 0.47 sec

$n = 2000$: 4.87 sec

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$\text{LMO}_C(x)$: requires only **first singular value/vectors**
 $(\sigma_1, U_1, V_1^\top)$

Full SVD

$n = 500$: 0.11 sec

$n = 1000$: 0.47 sec

$n = 2000$: 4.87 sec

Just $(\sigma_1, U_1, V_1^\top)$

$n = 500$: 0.0081 sec

$n = 1000$: 0.056 sec

$n = 2000$: 0.638 sec

Comparison with projections

Challenge: While **Projections** are analytically “well-behaved”

- Lipschitz continuous ($L = 1$)
- Firmly nonexpansive
- Fixed-point theory
- $\vdots$

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LMOs are *discontinuous*.

Operations are not interchangeable.

New theory and algorithms are often needed.

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A problem with multiple constraints

This line of work started with

$$\underset{x \in \bigcap_{i=1}^m C_i}{\text{minimize}} \quad f(x) \quad (*)$$

- ∇f is L_f -Lipschitz continuous (nonconvex)
- C_i are nonempty, compact, convex subsets of a real Hilbert space $\mathcal{H}$.

Classical projection-based approaches

$$\underset{x \in \bigcap_{i=1}^m C_i}{\text{minimize}} \quad f(x) \quad (*)$$

For appropriately-chosen stepsizes $(\gamma_k)_{k \in \mathbb{N}}$, $(*)$ can be solved via Forward-Backward, AKA projected gradient descent (PGD) using ∇f and $\text{proj}_{\bigcap_{i=1}^m C_i}$:

$$x_{k+1} = \text{proj}_{\bigcap_{i=1}^m C_i}(x_k - \gamma_k \nabla f(x_k))$$

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$$\sum_{i=1}^m [\text{Cost of } \text{proj}_{C_i}] \ll [\text{Cost of } \text{proj}_{\bigcap_{i=1}^m C_i}].$$

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Algorithmic “splitting” idea:



$$\text{proj}_{\bigcap_{i=1}^m C_i}$$



$$\text{proj}_{C_1}, \dots, \text{proj}_{C_m}$$

Classical projection-based approaches

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Projection-based splitting algorithms (dating back to the 1970s) e.g., Douglas-Rachford, Forward-Backward, ADMM, and many others solve $(*)$ using:

- First-order information on f (∇f or prox_f)
- $\text{proj}_{C_1}, \dots, \text{proj}_{C_m}$

Algorithmic “splitting” idea:



$$\text{proj}_{\bigcap_{i=1}^m C_i}$$



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A Frank-Wolfe setup for the splitting problem

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However, just like projections,

$$\sum_{i=1}^m [\text{Cost of } \text{LMO}_{C_i}] \ll [\text{Cost of } \text{LMO}_{\bigcap_{i=1}^m C_i}].$$



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Goal: Solve $(*)$ using basic algebraic operations and

- ∇f
- *Linear minimization oracles* for C_i :

$$\text{LMO}_{C_i}: x \mapsto y \in \underset{v \in C_i}{\operatorname{argmin}} \langle x \mid v \rangle$$

This is splitting, but with **LMOs** instead of **projection operators**.



Literature

Frank-Wolfe splitting algorithms when f is **convex**: [Gidel et al., '18], [Yurtsever et al., '19], [Silveti-Falls et al., '20], [Lan et al., '21], ...

Best convergence rate on $\max\{|f(x_k) - f^*|, \text{dist}_{\bigcap_{i=1}^m C_i}(x_k)\}$ is $\mathcal{O}(1/\sqrt{k})$

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With uniform convexity assumption: rate arbitrarily close to $\mathcal{O}(1/k)$.

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Today's talk: Frank-Wolfe splitting when f is **nonconvex**: [W., Pokutta, '25], [Halbey et al., '25], [Silveti-Falls, Molinari, W., '26]

Frank-Wolfe recap

Classical Frank-Wolfe algorithm
is for

$$\underset{x \in C}{\text{minimize}} \quad f(x). \quad (1)$$

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For $k \in \mathbb{N}$ and $\gamma_k \in [0, 1]$:

1. $v_k = \text{LMO}_C(\nabla f(x_k))$
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Guarantees $x_k \in C$.

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Stationarity criterion: *Frank-Wolfe gap* of (1) at x :

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$$G(x) = \max_{v \in C} \langle \nabla f(x) \mid x - v \rangle.$$

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Under a variety of schedules γ_t :

- $0 \leq \min_{0 \leq j \leq k} G(x_j) \leq Ck^{-1/2} \rightarrow 0$

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Requires LMO_C and ∇f .

Facts:

- If $x \in C$, $G(x) \geq 0$.
- $[x \in C \text{ and } G(x) = 0] \Leftrightarrow x$ is stationary for (1).

Under a variety of schedules γ_t :

- $0 \leq \min_{0 \leq j \leq k} G(x_j) \leq Ck^{-1/2} \rightarrow 0$
- Along the subsequence minimizing G , all cluster points are stationary points.
- [Bolte, C. Combettes, Pauwels, '23]:
 $(x_k)_{k \in \mathbb{N}}$ may not converge

Frank-Wolfe Splitting Algorithms

1. Background
2. History & Motivation
- 3. Approach**
4. Analysis
5. Conclusion

Product space reformulation

[Pierra, '84]: Reformulation of $(*)$ on $\mathcal{H}^m$: Let $\{\omega_i\}_{i=1}^m \subset]0, 1]$ s.t. $\sum_{i=1}^m \omega_i = 1$, let $A: \mathcal{H}^m \rightarrow \mathcal{H}: \mathbf{x} \mapsto \sum_{i=1}^m \omega_i x_i$, and set $D = \{(x_1, \dots, x_m) \mid (\forall i, j \in \{1, \dots, m\} \ x_i = x_j)\}$.

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$$x \in \underset{z \in \bigcap_{i=1}^m C_i}{\text{Argmin}} f(z) \quad \Leftrightarrow \quad (x, \dots, x) \in \underset{z=(z_1, \dots, z_m) \in D \cap \bigtimes_{i=1}^m C_i}{\text{Argmin}} f(Az)$$

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Reformulated Frank-Wolfe gaps are the same:

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Caution: This Frank-Wolfe gap can be negative!

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Setting of [S-F, M, W, '26+]

$$\underset{x \in \mathcal{C}}{\text{minimize}} f(x) + g(Tx) \quad (\star)$$

1. $\mathcal{C} \subset \mathbb{R}^n$ is nonempty, convex, and compact
2. $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and ∇f is $L_{\nabla f}$ -Lipschitz continuous (nonconvex) on $\mathcal{C}$

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4. $g: \mathbb{R}^m \rightarrow]-\infty, +\infty]$ is ρ -weakly convex and one of the following holds.
 - 4.1 $\mathcal{D} \subset \mathbb{R}^m$ is a nonempty closed convex set and $g(x) = \iota_{\mathcal{D}}(x) = \begin{cases} 0 & \text{if } x \in \mathcal{D} \\ +\infty & \text{if } x \notin \mathcal{D}, \end{cases}$
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The splitting problem $(\star)$ is a special case of $(\star)$ under Assumption 4.1 for a linear

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Today, we'll mostly focus on Assumption 4.1 instead of 4.2.

Setting of [S-F, M, W, '26+]

$$\underset{x \in \mathcal{C}}{\text{minimize}} \ f(x) + g(Tx) \quad (\star)$$

We assume we can compute:

- ∇f
- $\text{LMO}_{\mathcal{C}}$
- $\text{prox}_{\beta g}$ for $\beta \in]0, \rho^{-1}[$. (i.e., under Assumption 4.1, $\text{proj}_{\mathcal{D}}$;

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We'll be using the *Moreau envelope* of g for $\beta \in]0, \rho^{-1}[$:

$$g^{\beta}(x) = \min_{z \in \mathbb{R}^n} g(z) + \frac{1}{2\beta} \|x - z\|^2 \quad \Rightarrow \quad \nabla g^{\beta}(x) = \beta^{-1}(x - \text{prox}_{\beta g} x)$$

This avoids smoothing $g \circ T$.

Approach: a sequence of smoothed subproblems

$$\underset{x \in \mathcal{C}}{\text{minimize}} \ f(x) + g(Tx) \quad (\star)$$

For $\beta \in]0, \rho^{-1}[$, consider the smoothed problem:

$$\underset{x \in \mathcal{C}}{\text{minimize}} \ \Phi_\beta(x) := f(x) + g^\beta(Tx). \quad (\text{S})$$

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For the splitting problem $(*)$, the smoothed problem (S) becomes the relaxation,

$$\underset{\mathbf{x}=(x_1, \dots, x_m) \in \bigtimes_{i=1}^m \mathcal{C}_i}{\text{minimize}} \quad f(A\mathbf{x}) + \frac{1}{2\beta} \text{dist}_D^2(\mathbf{x}). \quad (R)$$

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Pseudocode: For a positive sequence $\beta_k \searrow 0$, consider

(A) Perform *one* Frank-Wolfe step for minimizing $\Phi_{\beta_k}(x) = f(x) + g^{\beta_k}(Tx)$ over $\mathcal{C}$.

(B) Decrease β_k , go to (A).

Algorithm

Algorithm: Frank-Wolfe with Moreau Envelope Smoothing (**Frames**)

Input: $x_0 \in \mathcal{C}$, step sizes $\{\gamma_k\}_{k \in \mathbb{N}} \subset]0, 1]$, smoothing schedule $\{\beta_k\}_{k \in \mathbb{N}} \subset]0, \rho^{-1}[$

For $k = 0, 1, 2, \dots$ **do**

1. Compute the gradient of the smoothed objective

$$\nabla \Phi_{\beta_k}(x_k) = \nabla f(x_k) + \frac{1}{\beta_k} T^* \left(T_{x_k} - \text{prox}_{\beta_k g}(T_{x_k}) \right).$$

2. Compute the LMO for this direction $s_k \in \arg \min_{s \in \mathcal{C}} \langle \nabla \Phi_{\beta_k}(x_k), s \rangle$.
3. Update the iterate $x_{k+1} = x_k + \gamma_k(s_k - x_k)$.

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Usually: For $0 < q < p < 1$, $p + q < 1$, $\gamma_k = \Theta((k+1)^{-p})$ and $\beta_k = \Theta((k+1)^{-q})$

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End for

Guarantees feasibility in $\mathcal{C}$: $(\forall k \in \mathbb{N}) \quad x_k \in \mathcal{C}$, but under 4.1, $T x_k \in \mathcal{D}$ not guaranteed.

Base Splitting Algorithm

Input: $x_0 \in \times_{i=1}^m C_i$, stepsizes $\{\gamma_k\}_{k \in \mathbb{N}} \subset]0, 1]$, smoothing $\{\beta_k\}_{k \in \mathbb{N}} \subset]0, +\infty[$

For $k = 0, 1, 2, \dots$ **do**

1. $(\nabla \Phi_{\beta_k}(x_k))^i = \nabla f(Ax_k) + \frac{1}{\beta_k}((x_k)^i - A(x_k))$

2. **For** $i = 1, 2, \dots, m$ **do**

$$s_k^i = \text{LMO}_{C_i}((\nabla \Phi_{\beta_k}(x_k))^i, s).$$

end for

3. $x_{k+1} = x_k + \gamma_k(s_k - x_k).$

End for

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Stationarity criterion: Frank-Wolfe gaps

Smoothed Frank-Wolfe gap

$$G^{\beta_k}(x_k) = \max_{s \in \mathcal{C}} \langle \nabla \Phi_{\beta_k}(x_k) \mid x - s \rangle$$

$x \in \mathcal{C}$ & $G^{\beta_k}(x) = 0 \Leftrightarrow x$ is a stationary point of minimize $\Phi_k(z)$
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- Under Assumption 4.1 ($g = \iota_{\mathcal{D}}$), set $\tilde{\mathcal{C}} = \mathcal{C} \cap T^{-1}(\mathcal{D})$

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Since $Tx_k \in \mathcal{D}$ (and $x \in \tilde{\mathcal{C}}$) not guaranteed, $\tilde{G}(x_k)$ could be negative!

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- Not this talk's focus: Under Assumption 4.2 (Lipschitz ρ -weakly convex g)

$$G(x; \xi) = \max_{s \in \mathcal{C}} \langle \nabla f(x) + T^* \xi \mid x - s \rangle \text{ for } \xi \in \partial g(Tx).$$

$x \in \mathcal{C}$, $\xi \in \partial g(Tx)$, & $G(x; \xi) = 0 \Leftrightarrow x$ is a stationary point of $(\star)$.

The base algorithm

If $\mathcal{H} = \mathbb{R}^n$, the algorithm [W., Pokutta '25] is a special case of “FraMes”

Split Conditional Gradient Algorithm [W., Pokutta, '25]

$T = \text{Id}$, $\mathcal{C} = \bigtimes_{i \in I} C_i$ and $\mathcal{D} = D$. Let $\beta_t = \mathcal{O}(1/\log(k+1))$ and $\gamma_k = \mathcal{O}(1/\sqrt{k+1})$.
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Q: Will this iteratively-smoothed F-W scheme work for $(*)$?

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If $\mathcal{H} = \mathbb{R}^n$, the algorithm [W., Pokutta '25] is a special case of “FraMes”

Split Conditional Gradient Algorithm [W., Pokutta, '25]

$T = \text{Id}$, $\mathcal{C} = \bigtimes_{i \in I} C_i$ and $\mathcal{D} = D$. Let $\beta_t = \mathcal{O}(1/\log(k+1))$ and $\gamma_k = \mathcal{O}(1/\sqrt{k+1})$. Set $G^{\beta_k}(\mathbf{x}_k) = \max_{\mathbf{v} \in \bigtimes_{i \in I} C_i} \langle \nabla \Phi_{\beta_k}(\mathbf{x}_k) \mid \mathbf{x}_k - \mathbf{v} \rangle$. Then

$$0 \leq \min_{0 \leq j \leq k} G^{\beta_j}(\mathbf{x}_j) \leq \mathcal{O}\left(\frac{\ln k}{\sqrt{k+1}}\right)$$

Also, $\text{dist}_D(\mathbf{x}_k) \rightarrow 0$. Along the subsequence with $G^{\beta_{k_j}}(\mathbf{x}_{k_j}) \rightarrow 0$, every cluster point yields a stationary point of $(*)$.

[Halbey et al., '25] Could remove the $\ln k$ factor with similar schedules β_k, γ_k .

Q: Will this iteratively-smoothed F-W scheme work for $(*)$? This analysis provides qualitative stationarity. What about a quantitative convergence rate on $\tilde{G}$?

Smoothed Frank-Wolfe gap convergence

Theorem (Silveti-Falls, Molinari, W.)

Let $p, q \in]0, 1[$ with $q < p$ and $p + q < 1$. Set step sizes $\gamma_k = (k + 1)^{-p}$ and smoothing schedule $\beta_k = (k + 1)^{-q}$. Then there exists $C_{p,q}$ such that

$$0 \leq \frac{1}{N} \sum_{k=0}^{N-1} G^{\beta_k}(x_k) \leq C_{p,q} N^{-\min\{p-q, 1-p-q\}}$$

Consequently, $0 \leq \min_{0 \leq k \leq N-1} G^{\beta_k}(x_k) \leq C_{p,q} N^{-\min\{p-q, 1-p-q\}}$. Along the subsequence with $G^{\beta_{k_j}}(x_{k_j}) \rightarrow 0$, every cluster point is a stationary point of $(\star)$.

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Stationarity arguments for (★)

Lemma (Silveti-Falls, Molinari, W.)

Under Assumption 4.1 ($g = \iota_{\mathcal{D}}$), for every $\beta > 0$,

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Stationarity argument: For $x \in \mathcal{C}$,

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So, if $\text{dist}_{\tilde{\mathcal{C}}}(x) \leq \kappa \text{dist}_{\mathcal{D}}(Tx)$,

$$\tilde{G}(x) \geq -L_f \kappa \text{dist}_{\mathcal{D}}(Tx) \quad \text{and} \quad -L_f \kappa \text{dist}_{\mathcal{D}}(Tx) + \frac{1}{\beta} \text{dist}_{\mathcal{D}}^2(Tx) \leq G^{\beta}(x)$$

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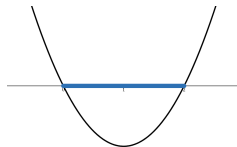
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Under some standard C.Q.'s (e.g., T surjective and $\text{relint}(\mathcal{C}) \cap \text{relint}(\mathcal{D}) \neq \emptyset$),

$$(\exists \kappa > 0) \quad (\forall x \in \mathcal{C}) \quad \text{dist}_{\mathcal{C} \cap T^{-1}(\mathcal{D})}(x) \leq \kappa \text{dist}_{\mathcal{D}}(Tx). \quad (\text{A})$$

For the splitting problem, $\bigcap_{i=1}^m \text{relint}(C_i) \neq \emptyset$ is sufficient for (A) [Bauschke et al., '99].

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Lemma (Silveti-Falls, Molinari, W.)

Let $x \in \mathcal{C}$ and $\beta > 0$. Suppose Assumption 4.1 and (A) hold. Then

$$\text{dist}_{\mathcal{D}}(Tx) \leq \frac{L_f \kappa \beta}{2} + \frac{1}{2} \sqrt{L_f \kappa^2 \beta^2 + 4\beta G^\beta(x)} = O\left(\max\{\beta, \sqrt{\beta G^\beta(x)}\}\right)$$

and

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- Characterizes nonsmooth stationarity in terms of smoothing parameter β and smoothed F-W gap G^β
- Algorithm independent bounds!
- Shows that for previous works [W., Pokutta, '25], [Halbey et al., '25], both feasibility $\text{dist}_{\mathcal{D}}(Tx_k)$ and stationarity $|\tilde{G}(x_k)|$ converge like $\mathcal{O}(1/\log k)$.

What this means for the algorithm

Theorem (Silveti-Falls, Molinari, W.)

Suppose Assumption 4.1 and (A) holds, and let $\{x_k\}_{n \in \mathbb{N}}$ be generated by the algorithm with $\gamma_k = (k+1)^{-1/2}$ and $\beta_k = \beta_0(k+1)^{-1/4}$.

For $N \geq 2$, let $k_N^* \in \underset{\lfloor N/2 \rfloor \leq k \leq N-1}{\operatorname{argmin}} G^{\beta_k}(x_k)$. Then

$$\operatorname{dist}_{\mathcal{D}}(Tx_{k_N^*}) = \mathcal{O}(N^{-1/4}) \quad \text{and} \quad |\tilde{G}(x_{k_N^*})| = \mathcal{O}(N^{-1/4}).$$

	Convex objective	Nonconvex objective
One constraint	$\mathcal{O}(1/k)$	$\mathcal{O}(1/\sqrt{k})^*$
Multiple constraints	$\mathcal{O}(1/\sqrt{k})^{**}$	[S-F, M, W '26]: $\mathcal{O}(1/k^{1/4})$

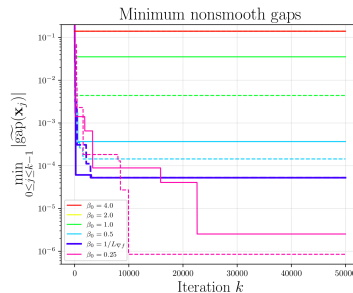
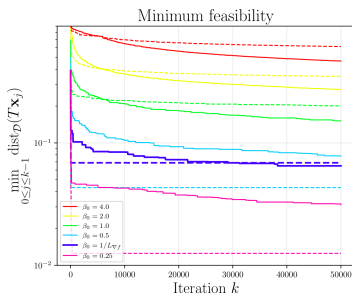
Extra assumptions can increase the rate, e.g.:

*: [Díaz Millán et al., '25];

** : [Wang, Pong, W., '26]

Other results

- Analogous results with g nonsmooth, weakly-convex, & Lipschitz continuous
- Inconsistent problems ($\mathcal{C} \cap T^{-1}\mathcal{D} = \emptyset$)
- Numerical experiments



sold lines – logarithmic smoothing;
Dashed lines – proposed smoothing schedules

Frank-Wolfe Splitting Algorithms

1. Background
2. History & Motivation
3. Approach
4. Analysis
- 5. Conclusion**

Next steps

Input: $x_0 \in \times_{i=1}^m C_i$, stepsizes $\{\gamma_k\}_{k \in \mathbb{N}} \subset]0, 1]$,
smoothing $\{\beta_k\}_{k \in \mathbb{N}} \subset]0, +\infty[$

For $k = 0, 1, 2, \dots$ **do**

1. $\nabla(\Phi_{\beta_k}(x_k))^i = \nabla f(Ax_k) + \frac{1}{\beta_k}((x_k)^i - A(x_k))$

2. **For** $i = 1, 2, \dots, m$ **do**

$$s_k^i = \text{LMO}_{C_i}((\nabla \Phi_{\beta_k}(x_k))^i, s).$$

End for

3. $x_{k+1} = x_k + \gamma_k(s_k - x_k).$

End for

To be announced:

- Rate improvement to $\mathcal{O}(1/k^{1/3})$

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If $i \in I_t$

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- Adaptive smoothing

Conclusion

“Moreau smoothing + one F-W step” works in more settings than splitting.

Contact: `woodstzc[at]jmu.edu`

Preprint:

Frank-Wolfe with Moreau Envelope Smoothing for
Nonsmooth Nonconvex Problems,

Antonio Silveti-Falls, Cesare Molinari, and Zev Woodstock,
arXiv [arXiv:2606.00853](https://arxiv.org/abs/2606.00853)



Thank you for your attention!

References

-  K. Asgari, M. J. Neely, Nonsmooth projection-free optimization with functional constraints
Comput. Optim. Appl., vol. 89 (3), pp. 927–975, 2024.
-  H. H. Bauschke, J. M. Borwein, W. Li, Strong conical hull intersection property, bounded linear regularity, Jameson's property (G), and error bounds in convex optimization
Math. Program., Ser. A, vol. 86 (4) pp. 135–160, 1999.
-  M. Besançon, M. Carderera, and S. Pokutta, FrankWolfe. jl: A high-performance and flexible toolbox for Frank–Wolfe algorithms and conditional gradients
INFORMS J. Comput., vol. 34 (5), pp. 2611–2620, 2022.
-  J. Bolte, C. W. Combettes, and E. Pauwels, The iterates of the Frank–Wolfe algorithm may not converge
Math. Oper. Res., vol. 49 (4), pp. 2565–2578, 2024.
-  C. W. Combettes and S. Pokutta, Complexity of linear minimization and projection on some sets
Oper. Res. Lett., vol. 49, no. 4, pp. 565–571, 2021
-  R. Díaz Millán, O. P. Ferreira, and J. Ugon, Frank–Wolfe algorithms for piecewise star-convex functions with a nonsmooth difference-of-convex structure
arXiv preprint 2308.16444, rev. 2026.

References

-  D. Garber, A. Kaplan, and S. Sabach, Improved complexities of conditional gradient-type methods with applications to robust matrix recovery problems, *Math. Program.*, vol. 186, pp. 185–208, 2021.
-  G. Gidel, F. Pedregosa, and S. Lacoste-Julien, Frank-Wolfe splitting via augmented Lagrangian Method, *Proc. AISTATS*, pp. 1456–1465, 2018.
-  J. Halbey, S. Rakotomandimby, M. Besançon, S. Designolle, and S. Pokutta, Efficient quadratic corrections for Frank-Wolfe algorithms, *Proc. NeurIPS 38*, 2025.
-  T. Hu, Robustness of the Frank-Wolfe method under inexact oracles and the cost of linear minimization
arXiv preprint 2601.11548, 2026.
-  M. Jaggi, Revisiting Frank-Wolfe: Projection-free sparse convex optimization, *Proc. 30th International Conference on Machine Learning*, in PMLR, vol. 28 (1), pp. 427–435, 2013.

References

-  G. Lan, E. Romeijn, and Z. Zhou, Conditional gradient methods for convex optimization with general affine and nonlinear constraints,
SIAM J. Optim., vol. 31, no. 3, pp. 2307–2339, 2021.
-  G. Pierra, Decomposition through formalization in a product space,
Math. Program., vol. 28, pp. 96–115, 1984.
-  A. Silveti-Falls, C. Molinari and J. Fadili, Inexact and stochastic generalized conditional gradient with augmented Lagrangian and proximal step,
J. Nonsmooth Anal. Optim., vol. 2, 2021.
-  X. Wang, T. K. Pong, and Z. Woodstock, A conditional-gradient-based single-loop augmented Lagrangian method for inequality constrained optimization,
arXiv preprint 2605.22539, 2026.
-  Z. Woodstock, High-precision linear minimization is no slower than projection,
Optim. Lett., vol. 20, pp. 909–915, 2026.
-  Z. Woodstock and S. Pokutta, Splitting the conditional gradient algorithm,
SIAM J. Optim., vol. 35 (1), pp. 347–368, 2025.

References



A. Yurtsever, O. Fercoq, F. Locatello, and V. Cevher, A conditional gradient framework for composite convex minimization with applications to semidefinite programming
Proc. ICML, 2018.



A. Yurtsever, O. Fercoq, and V. Cevher, A conditional-gradient-based augmented Lagrangian framework
Proc. ICML, pp. 7272–7281, 2019.